

The motivation of the paper is to present a framework and fairness metric for rethinking the Web architecture for affordability and inclusion. The UN Broadband Commission for Sustainable Development has set the target for affordable broadband services to be less than 2% of the gross national income, which 94 developing countries do not meet. Increasing web complexity makes the internet even less affordable. The authors present Affordable Web for All (AW4A) framework to tackle the issue, which relies on a new fairness metric Price Adjusted Web access (PAW). The framework aims to maximize the webpage quality while keeping it under the given target size. The analysis shows that reducing the average web page size by 1.5x allows up to 14% of the countries to meet affordability targets.

The authors observe that median web page size has increased by 28% in the last five years and variations in popularity of webpages across regions leads to variations in average web page size. In general, web pages accessed from developing countries tend to be larger in size than those accessed from developed countries. The authors use the average mobile broadband price, average web page size and global average web page size to compute the PAW index and the expected number of web accesses a user can afford before running out of the 2% monthly target price.

The authors find that JS and images contribute to the most bytes (23%-47% and 24% to 58%) respectively. Removing these resources can lead to an 15x reduction in page size. However, removing these elements can negatively impact the webpage quality. The paper reports a user study to understand the impact of quality-access trade-off. The average monthly broadband expenditure of the users in the study was 4.1%. Optimizing the webpages for various levels of reduction in page size showed that all pages remained usable by 1.5x reduction while only some remained functional beyond that. For some websites, 1.25% is achievable by simply switching to webp format.

The framework is designed with two primary goals; the framework should be able to adapt the complexity of webpages based on mobile broadband prices and it should preserve end to end TLS encryption. Website operator consent is also required. For the first goal, AW4A explicitly accounts for affordability in its design. For the second, it transcodes websites on the server end rather than the proxy end. It provides a developer API as well as a switch for the user to enable the desired percentage of savings. The framework benefits all involved parties, it offers the website operators an opportunity to offer their services at reduced quality to make them more accessible, it benefits users by giving them the choice to reduce the quality as needed, and it makes it attractive to mobile network operators to offer smaller data plans at reduced costs.

The work aims to solve a pressing issue by presenting a comprehensive framework, however, there is one issue that requires attention. Informed choice requires educating the user. An opt-in feature like this (a switch for the user as the paper mentions) could exist on a device or a website without the user ever learning about it. The study does not appear to be considering the literacy level of the user. A user may unknowingly enable the feature and in case a website breaks, they may not be able to figure out how to fix it.